Processing a Real GC-MS/FID Sample

The problem

A fuel or process sample leaves the instrument as three separate things: an FID peak table (quantitative, but anonymous), an MSD peak table (identified, but on a slightly different time axis), and a library-search export that names the MSD peaks. A client wants one table of compounds with amounts, plus a summary of chemical families and elements.

`fidms` replaces the old "Automated Polyarc" Excel/VBA macro for this job. This vignette follows one real sample, `D319`, end to end, first in code and then through the local Shiny app. Along the way it points out each place where the analyst has to supply something the instrument files cannot know.

What goes in

The three exports `fidms` reads are bundled with this vignette as CSV files, copied from the Agilent `D319.D` data directory. The raw `data.ms` scan file in that directory enables the optional hydrocarbon-class analysis; it is not bundled, so that section only runs where the `.D` directory is available.

```
fid_csv    <- "FIDpeaks.csv"
msd_csv    <- "MSDpeaks.csv"
search_csv <- "MSDsearch.csv"

file.exists(c(fid_csv, msd_csv, search_csv))
#> [1] TRUE TRUE TRUE
```

Only the three CSVs are required; the raw scans are additive.

Sample properties only the analyst knows

Several numbers feed directly into the reported weight percents, and none of them are in the instrument files:

Property	Why it is needed
Sample number, name, project code, analyst(s)	Report headers and output file name
Density (g/mL)	Converts μg/mL to weight percent
Methanol and water (wt. %)	Measured separately (e.g. Karl Fischer); reported alongside the GC result so totals can approach 100
Dilution factor	Corrects amounts for any dilution before injection

```
info <- fidms:::.sample_info(
  sample_id       = "D319",
  sample_name     = "Demo fuel sample",
  project_code    = "DEMO",
  analyst         = "PB",
  density         = 0.8519,
  methanol_wt_pct = 0,
  water_wt_pct    = 53 / 1e4, # 53 ppm = 0.0053 wt. %
  dilution        = 1
)
info$analyst
#> [1] "PB"
```

The metadata constructor is internal, because the Shiny app is its main caller; in code you build it directly. Multiple analysts are joined with a slash. The density and water content are the measured values for this sample; water is entered in weight percent, so 53 ppm becomes 0.0053.

Processing in code

Alignment and identification

`prepare_peaks()` is the first half of the pipeline. It parses the files, finds the time offset between the FID and MSD detectors, matches peaks across them, attaches library identities, merges split peaks and fills in formulae for unidentified peaks by interpolation.

```
prep <- prepare_peaks(fid_csv, msd_csv, search_csv)
c(peaks = nrow(prep$peaks), shift_min = prep$shift)
```

```
#>     peaks shift_min
#>   650.000     0.009
table(prep$peaks$origin)
#>
#> extrapolated interpolated          msd
#>           90          247          313
```

The search found an MS-to-FID offset of 0.009 minutes, which is small. Most peaks align to an MSD peak; the few `interpolated` and `extrapolated` ones have no library identity, so their formulae are estimated from neighbours. Those are the rows to treat with the most caution in a client report.

Optional: hydrocarbon classes from raw scans

When the raw `data.ms` is available, the ASTM D2425 class-ion deconvolution can assign hydrocarbon classes to peaks the library could not name, instead of leaving them to interpolation. These helpers are internal; the Shiny app calls them for you whenever you give it a `.D` directory.

```
scan_matrix  <- fidms:::.read_ms_scan_matrix(d_dir)
class_traces <- fidms:::.build_class_ion_traces(scan_matrix)

prep_cls <- prepare_peaks(
  fid_csv, msd_csv, search_csv,
  class_traces = class_traces,
  scan_matrix  = scan_matrix
)
table(prep_cls$peaks$assignment_tier, useNA = "ifany")
#>
#>   A   B   C   D   M
#> 237  71 256   9  77
```

The assignment tiers record how much evidence supports each peak (library hit, class-ion solve, mixed region, or none). Reading the raw scan file takes several seconds, which is why this step is opt-in.

Quantitation

Calibration is the second half. The default `style = "slope"` uses the packaged calibration table for compounds with targets and a Polyarc relative response for everything else. Dilution and density come from the sample metadata.

```
peaks <- calibrate(
  prep_cls$peaks,
  style    = "slope",
  dilution = info$dilution,
  density  = info$density
)
res <- summarize_report(peaks, info, shift = prep_cls$shift)
res$totals[c("gc_recovery", "total")]
#> $gc_recovery
#> [1] 88.90453
#>
#> $total
#> [1] 88.90983
```

GC recovery is the sum of the quantified, non-solvent compounds in weight percent. A value well below 100 after adding methanol and water would suggest undetected material, so the number is worth comparing with expectations for the sample type rather than accepting automatically.

Internal-standard quantitation

If the sample contains a compound that can serve as an internal reference, `style = "internal"` normalises the sample to 100% against it. Choosing that peak is a judgement call, so `fidms` proposes one and checks its quality.

```
ref <- fidms:::.suggest_reference_peak(prep_cls$peaks)
ref[c("peak_index", "name", "area_percent", "score", "approved")]
#> $peak_index
#> [1] 495
#>
#> $name
#> [1] "9-Octadecenoic acid, methyl ester, (E)-"
#>
#> $area_percent
#> [1] 3.881259
#>
#> $score
```

```
#> [1] 97.56
#>
#> $approved
#> [1] TRUE
```

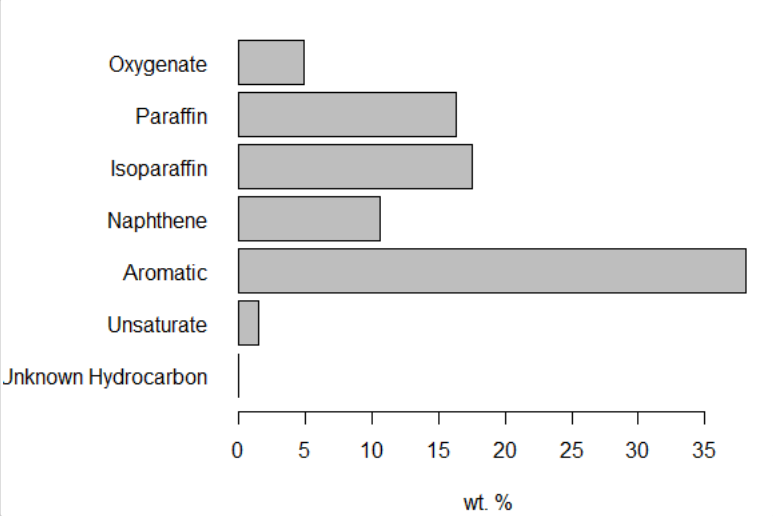
Here the suggestion is the largest peak that passes the quality check (a good library match and a reasonable share of the total area). `approved` is `TRUE` when the largest peak itself qualified; otherwise the next best qualifying peak is offered. The analyst may override it by choosing any other peak index.

```
peaks_is <- calibrate(
  prep_cls$peaks,
  style          = "internal",
  reference_peak = ref$peak_index,
  dilution       = info$dilution,
  density        = info$density
)
res_is <- summarize_report(peaks_is, info, shift = prep_cls$shift)
round(res_is$totals$gc_recovery, 1)
#> [1] 88
```

The internal-standard recovery differs from the slope-style result, because amounts are now scaled relative to the reference peak rather than from the calibration curves. Here the reference is a library-identified methyl ester making up a few percent of the total area, so the result depends on how well that one peak represents the sample. Comparing the two recoveries is a useful sanity check on the choice of reference.

Reading the summary

```
fam <- res$families
fam <- fam[fam$wt_pct > 0, ]
op <- par(mar = c(4, 9, 1, 1))
barplot(rev(fam$wt_pct), names.arg = rev(fam$compound_type), horiz = TRUE,
        las = 1, xlab = "wt. %")
```




```
par(op)
```

The sample is dominated by aromatics and the hydrocarbon families, with a smaller oxygenate fraction. Families are assigned programmatically from formulae and names, so the compound types are indicative and, as the report notes state, not a substitute for a standard PIONA analysis.

The per-compound table that the client sees is built by `total_compound_table()`:

```
tc <- total_compound_table(res, include_class_ion = TRUE)
head(tc[, c("Peak No.", "Compound Name", "Retention Time",
            "Area (FID)", "Amount (ug/mL)")], 8)
#> # A tibble: 8 × 5
#>   `Peak No.` `Compound Name`      `Retention Time` `Area (FID)` `Amount (ug/mL)`
#>   <chr>      <chr>                <chr>                  <dbl>            <dbl>
#> 1 1          Cyclopropene         1.327                  3898.             5.82
#> 2 2          Unknown, Interpolat… 1.377                 11609.            17.9
#> 3 3          Butane               1.404                 71176.            116.
#> 4 4          Butane, 2-methyl-    1.496                122781.            198.
#> 5 5          Pentane              1.544                206784.            334.
#> 6 6          <NA>                 1.596                  7822.             NA
```

```
#> 7 7          Methylene chloride   1.645              27526903.        261396.
#> 8 8          Pentane, 2-methyl-   1.757                674695.          1084.
```

Unmatched peaks have no compound name, and combined peaks print the sentinel `"Combined FID Peaks"` instead of a retention time, which is why that column is character.

Writing the reports

Both writers take the summarised result. The Excel report has a Total Compound sheet and a Summary sheet.

```
xlsx <- write_excel_report(res, file.path(tempdir(), "DEMO_D319_GC-MS-FID.xlsx"))
openxlsx::getSheetNames(xlsx)
#> [1] "D319 Total Compound Report" "D319 Summary Report"
```

The PDF report is rendered from a Quarto template, so it needs Quarto and a LaTeX engine installed. The chunk below is not evaluated when the vignette is built; the image after it is the first page of the report produced from this sample.

```
write_pdf_report(res, "DEMO_D319_GC-MS-FID.pdf")
```

GC-MS/FID Report

Total Compound Report

Sample ID	D319	Density	0.8519 g/mL	Aligned Peaks	313
Sample Name	Demo fuel sample	Methanol	0.0 wt. %	Unique Peaks	9
Project Code	DEMO	Water	0.0 wt. %	Total Peaks	650
Report Date	2026-10-08	GC Recovery	88.9 wt. %	Interpolated Peaks	9
Analyst	PB	Total:	88.9 wt. %	MS/FID Offset Time	0.009 min

Peak No.	Compound Name	Retention Time	Compound Formula	Compound Molar Mass	Area (FID)	Amount (ug/mL)	Target or Relative
1	Cyclopropene	1.327	C3 H4	40.065	3897.65	5.820	Relative
2	Unknown, Interpolated	1.377	C4H7	55.100	11608.98	17.881	Relative
3	Butane	1.404	C4 H10	58.124	71176.50	115.645	Relative
4	Butane, 2-methyl-	1.496	C5 H12	72.151	122780.54	198.106	Relative
5	Pentane	1.544	C5 H12	72.151	206783.79	333.645	Relative
6	NA	1.596	NA	NA	7822.48	NA	Relative
7	Methylene chloride	1.645	C H2 Cl2	84.927	27526903.05	61395.968	Relative
8	Pentane, 2-methyl-	1.757	C6 H14	86.178	674694.54	1083.548	Relative
9	Pentane, 3-methyl-	1.826	C6 H14	86.178	477251.74	766.458	Relative
10	n-Hexane	1.909	C6 H14	86.178	1010761.57	1623.266	Relative
11	NA	2.063	NA	NA	14846.69	NA	Relative
12	Cyclopentane, methyl-	2.118	C6 H12	84.162	887696.57	1392.275	Relative
13	Unknown, Interpolated	2.236	C6H14	86.178	1003.32	1.611	Relative
14	Pentane, 3,3-dimethyl-	2.337	C7 H16	100.205	13490.22	21.593	Relative
15	Cyclohexane	2.424	C6 H12	84.162	1283625.02	2013.255	Target
16	NA	2.468	NA	NA	260544.45	NA	Relative
17	Hexane, 3-methyl-	2.529	C7 H16	100.205	787882.10	1261.097	Relative
18	Cyclopentane, 1,3-dimethyl-	2.653	C7 H14	98.189	337824.55	529.849	Relative
19	NA	2.689	NA	NA	271962.36	NA	Relative
20	Cyclopentane, 1,2-dimethyl-	2.724	C7 H14	98.189	549901.09	862.472	Relative
21	Heptane	2.835	C7 H16	100.205	1656632.16	6974.479	Target
22 - 22	Unknown, Extrapolated Combined FID Peaks	Combined FID Peaks	C8H17	113.224	7166.27	8.445	Relative
24	Pentane, 2,2,4-trimethyl-	3.156	C8 H18	114.232	27280.87	32.433	Relative
25	Cyclohexane, methyl-	3.259	C7 H14	98.189	2662675.03	3983.161	Target
26	NA	3.332	NA	NA	110398.44	NA	Relative
27	Hexane, 2,4-dimethyl-	3.369	C8 H18	114.232	162057.19	184.825	Relative
28	Cyclopentane, ethyl-	3.439	C7 H14	98.189	408552.22	457.727	Relative
29	Cyclopentane, 1,2,4-trimethyl-	3.570	C8 H16	112.216	377513.40	422.952	Relative
30	Unknown, Interpolated	3.665	C8H16	112.216	2007.25	2.249	Relative
31	Cyclopentane, 1,2,3-trimethyl-, (1.alpha.,2.alpha.,3.beta.)-	3.729	C8 H16	112.216	361890.81	405.449	Relative
32	Pentane, 2,3,3-trimethyl-	3.885	C8 H18	114.232	15871.88	18.102	Relative
33	Hexane, 2,3-dimethyl-	3.979	C8 H18	114.232	220933.75	251.973	Relative
34	1,3,5-Cycloheptatriene	4.155	C7 H8	92.141	3701242.18	8891.313	Relative
35	NA	4.223	NA	NA	113641.75	NA	Relative
36	Heptane, 3-methyl-	4.328	C8 H18	114.232	832403.88	949.348	Relative
37	Cyclopentane, 1,2,4-trimethyl-	4.399	C8 H16	112.216	217592.18	243.782	Relative
38	Cyclohexane, 1,3-dimethyl-, cis-	4.543	C8 H16	112.216	1157490.05	1296.808	Relative
39	Cyclohexane, 1,4-dimethyl-	4.621	C8 H16	112.216	479445.11	537.152	Relative

First page of the generated PDF: sample header and QC counts, followed by the Total Compound table.

The header block repeats the sample metadata and the headline numbers (GC recovery, peak counts, MS/FID offset), so a reader can check the report against the inputs without opening anything else. Compounds without a library identity show as `NA`. The remaining pages continue the compound table and end with the Summary Report.

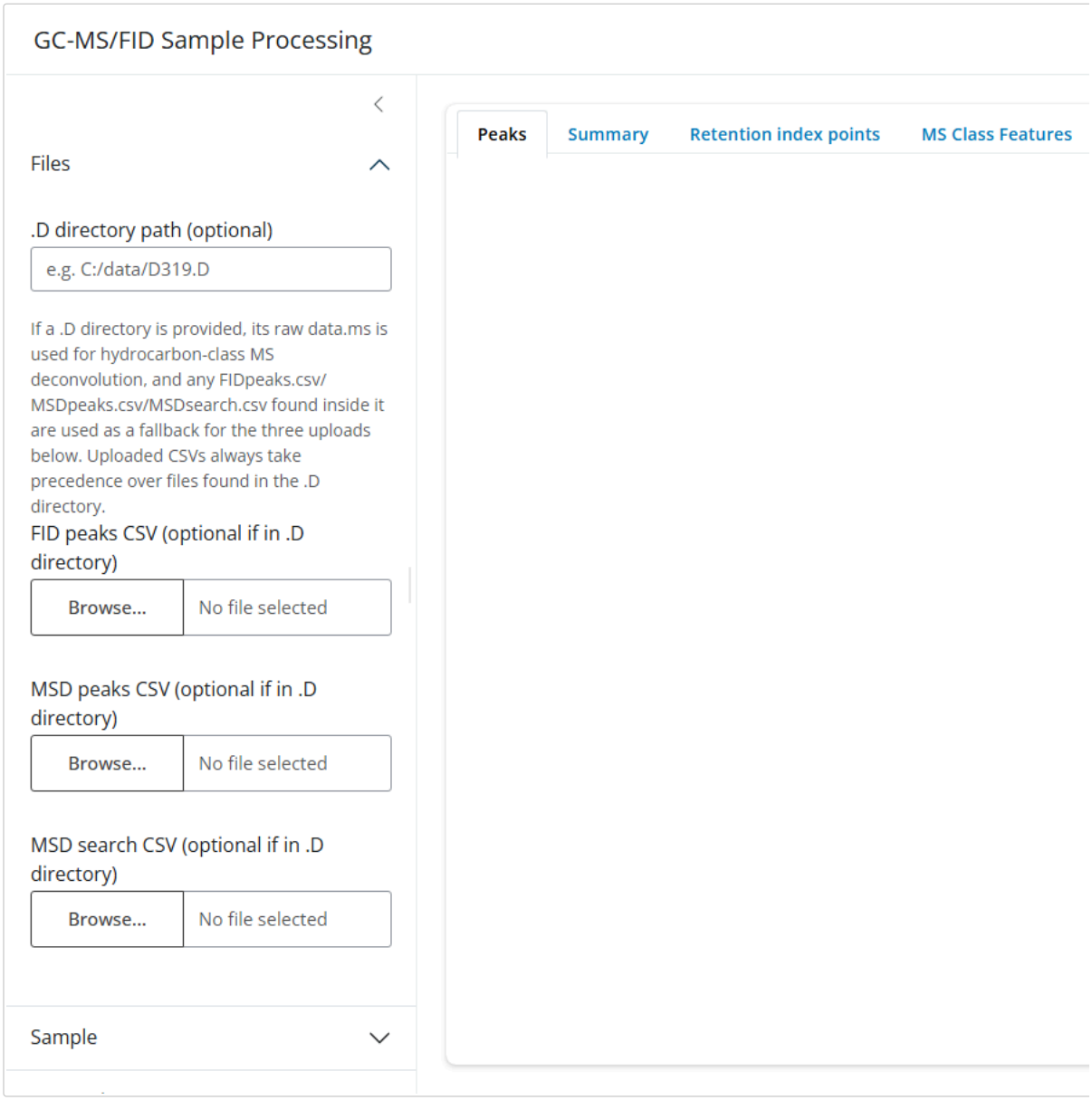
The same workflow in the Shiny app

Everything above is available through a local Shiny app. It runs on your machine and is not meant for deployment, so client data stays local.

```
fidms::run_app()
```

The app needs `shiny`, `bslib` and `DT`. The screenshots below were captured from a real run on `D319.D`.

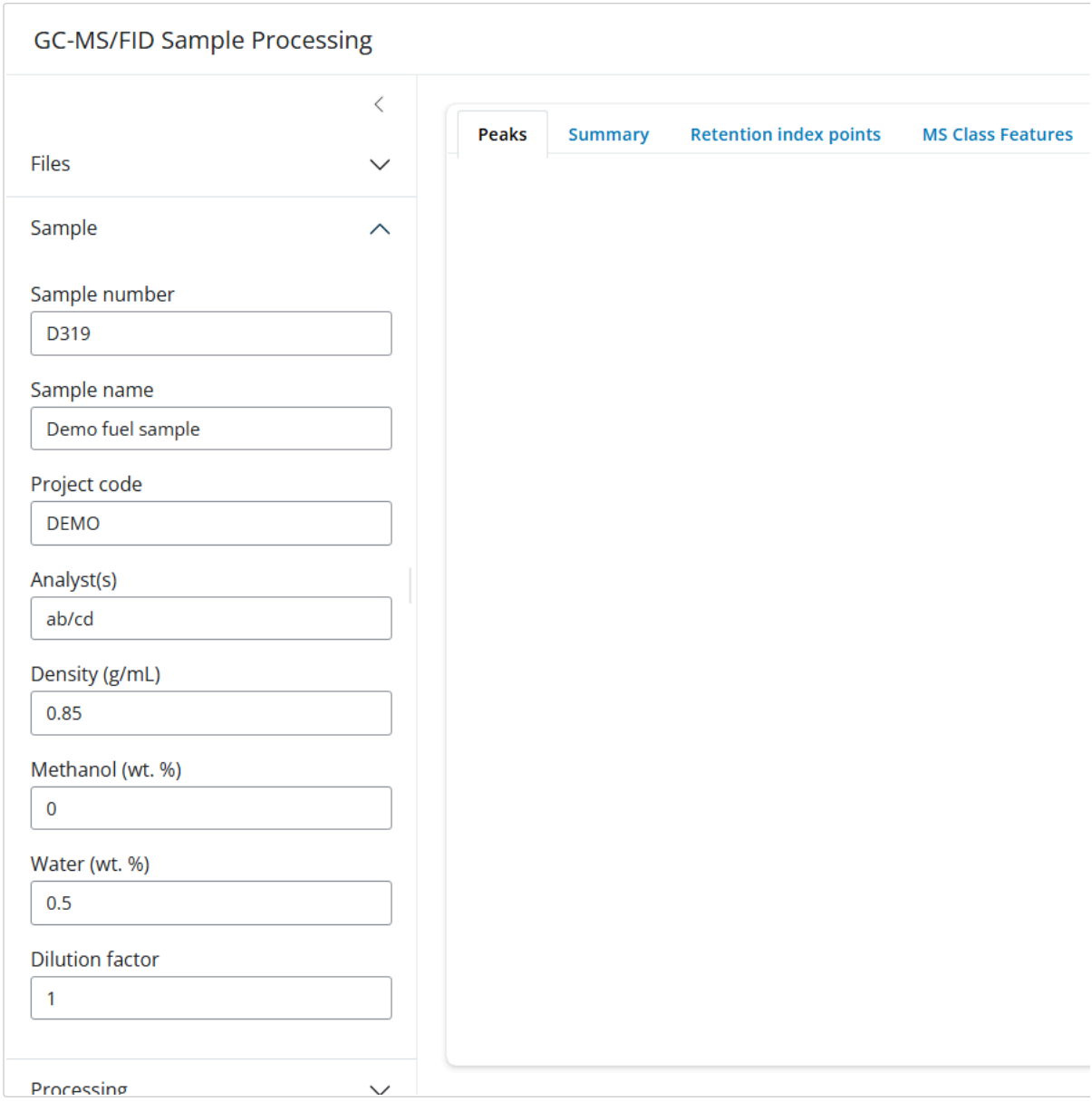
1. Point the app at the data



The app on start-up: the Files panel accepts a `.D` directory and/or uploads of the three CSVs.

Enter the `.D` directory and the app uses the CSVs found inside it, and also reads `data.ms` for the hydrocarbon-class analysis. Uploaded CSVs take precedence over files in the directory.

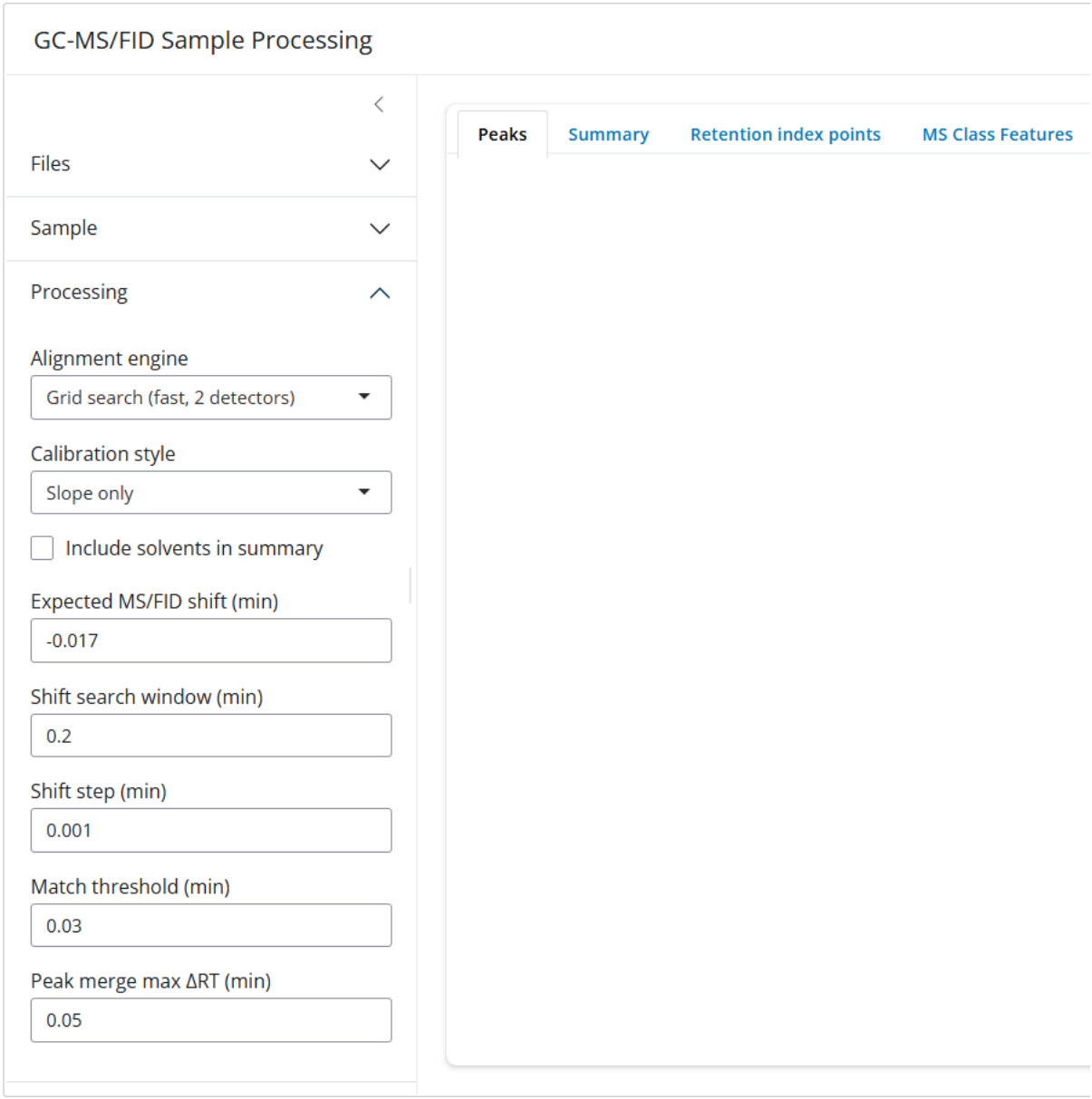
2. Enter the sample properties



The Sample panel holds the metadata the instrument files do not contain.

This is the same information as the metadata object in code. The sample number is required; processing does not start without it.

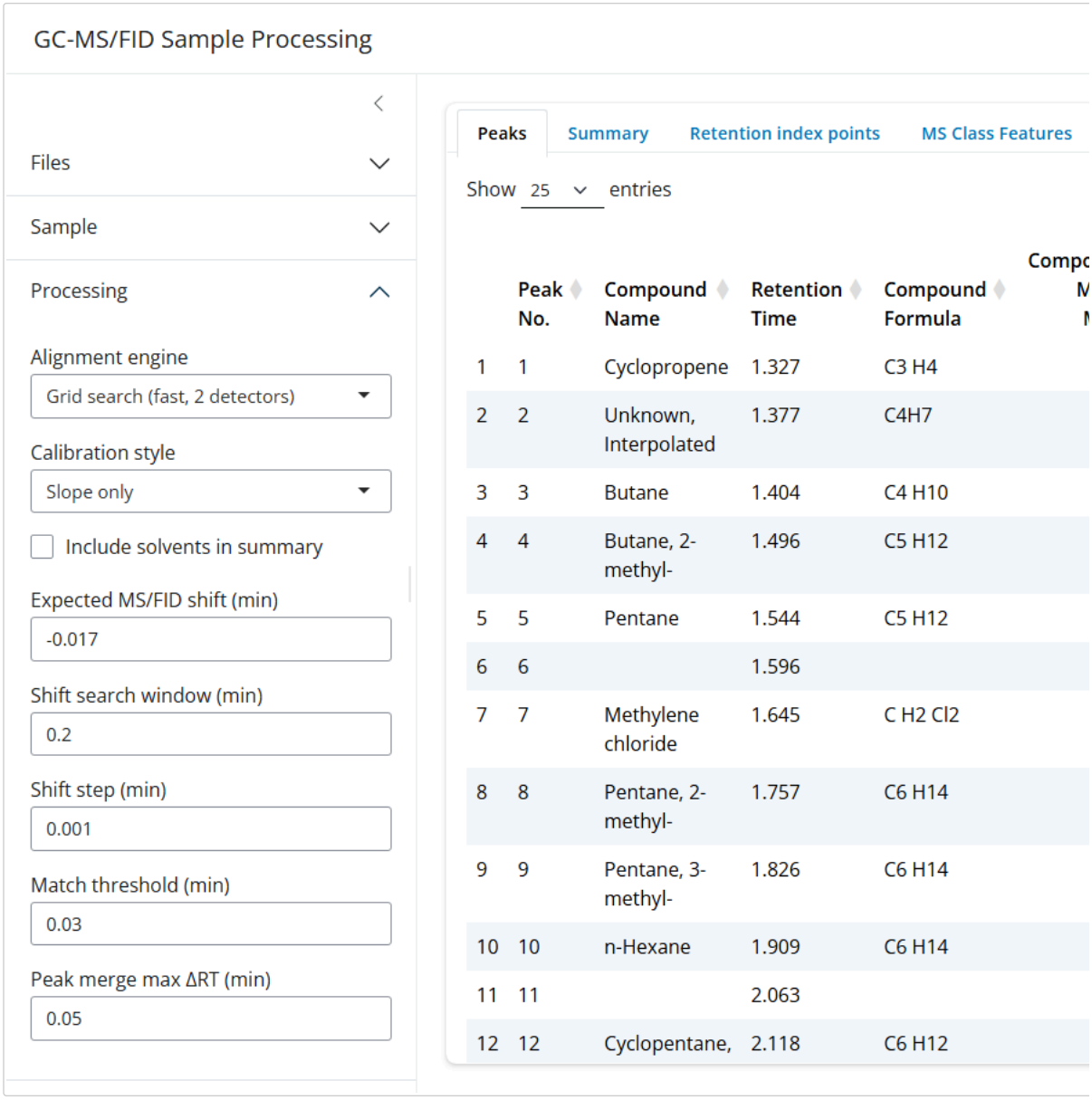
3. Choose processing options



The Processing panel: alignment engine, calibration style and shift-search parameters.

The defaults reproduce the macro's behaviour. The alignment engine (`grid` or `gcalignr`) and the shift parameters correspond to `fidms_params()`. The two engines are not expected to agree exactly, and the production default is still an open question. Choosing **Internal standard** adds a panel showing the suggested reference peak, which you can override by selecting a row in the Peaks tab.

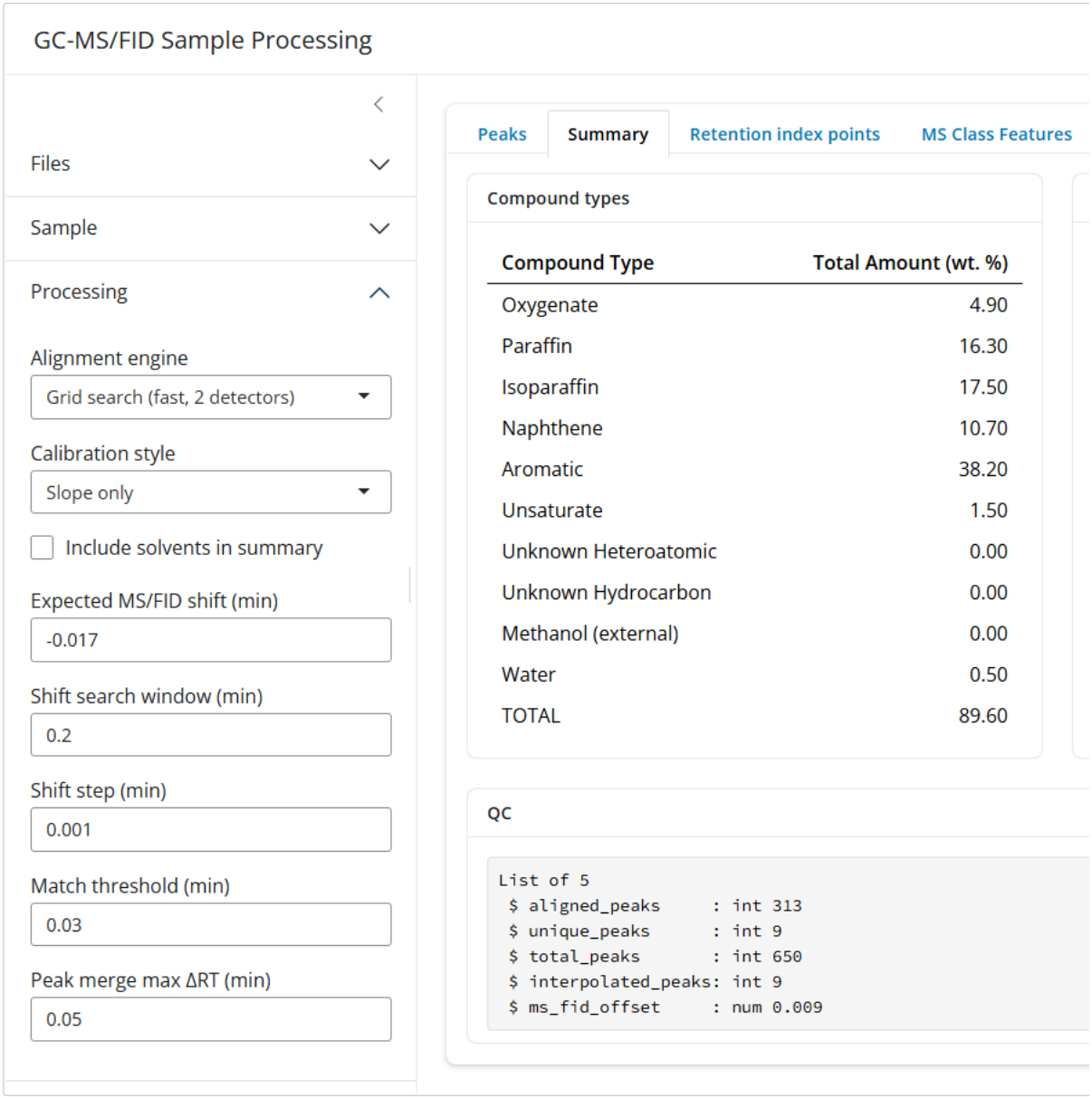
4. Process and review the peaks



The Peaks tab after pressing Process sample.

Pressing **Process sample** runs the same steps as the code above. The Peaks table is the Total Compound table; with a `.D` directory it also carries the class-ion columns to the right.

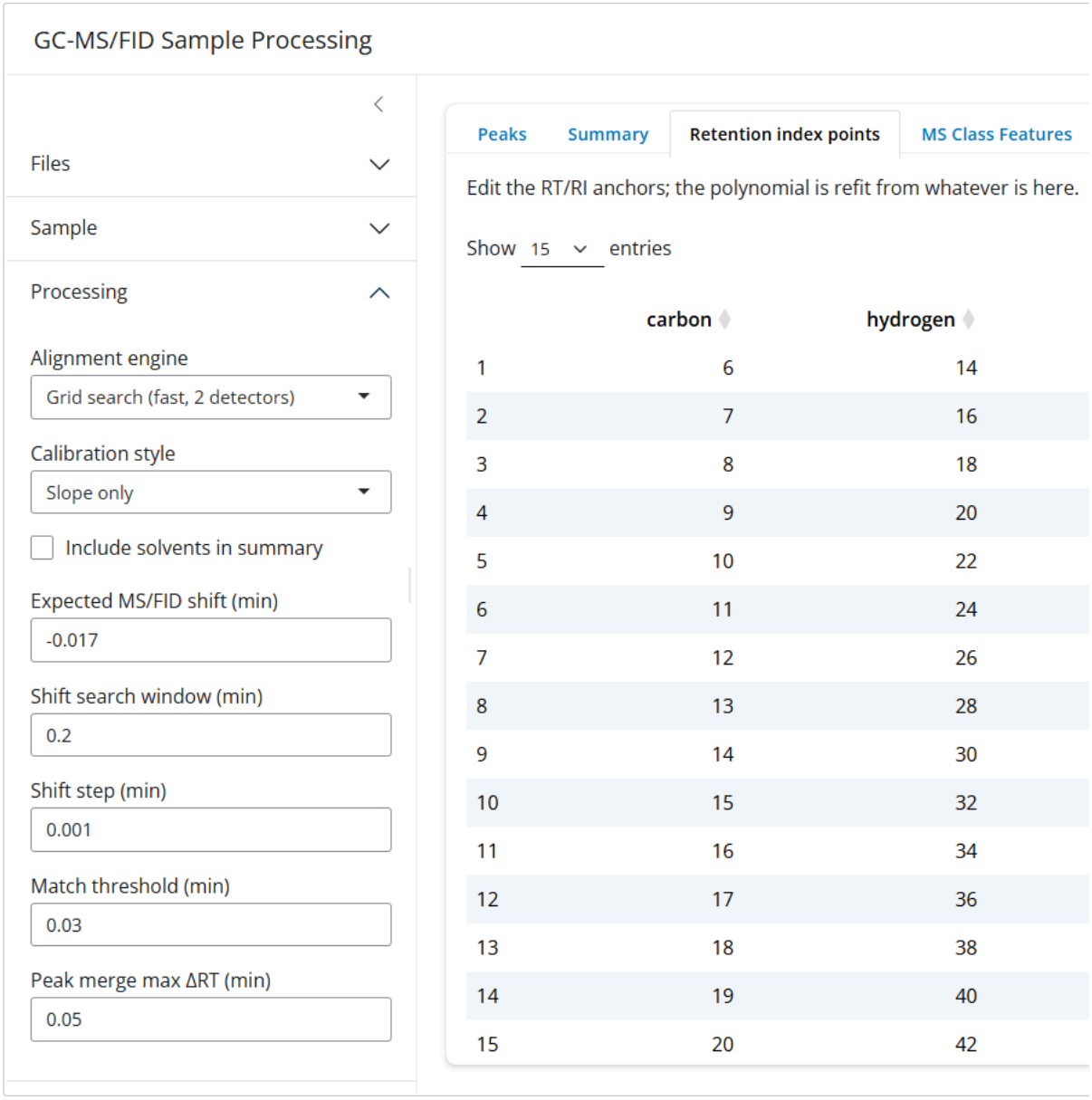
5. Check the summary



The Summary tab: compound types, elements and QC counts.

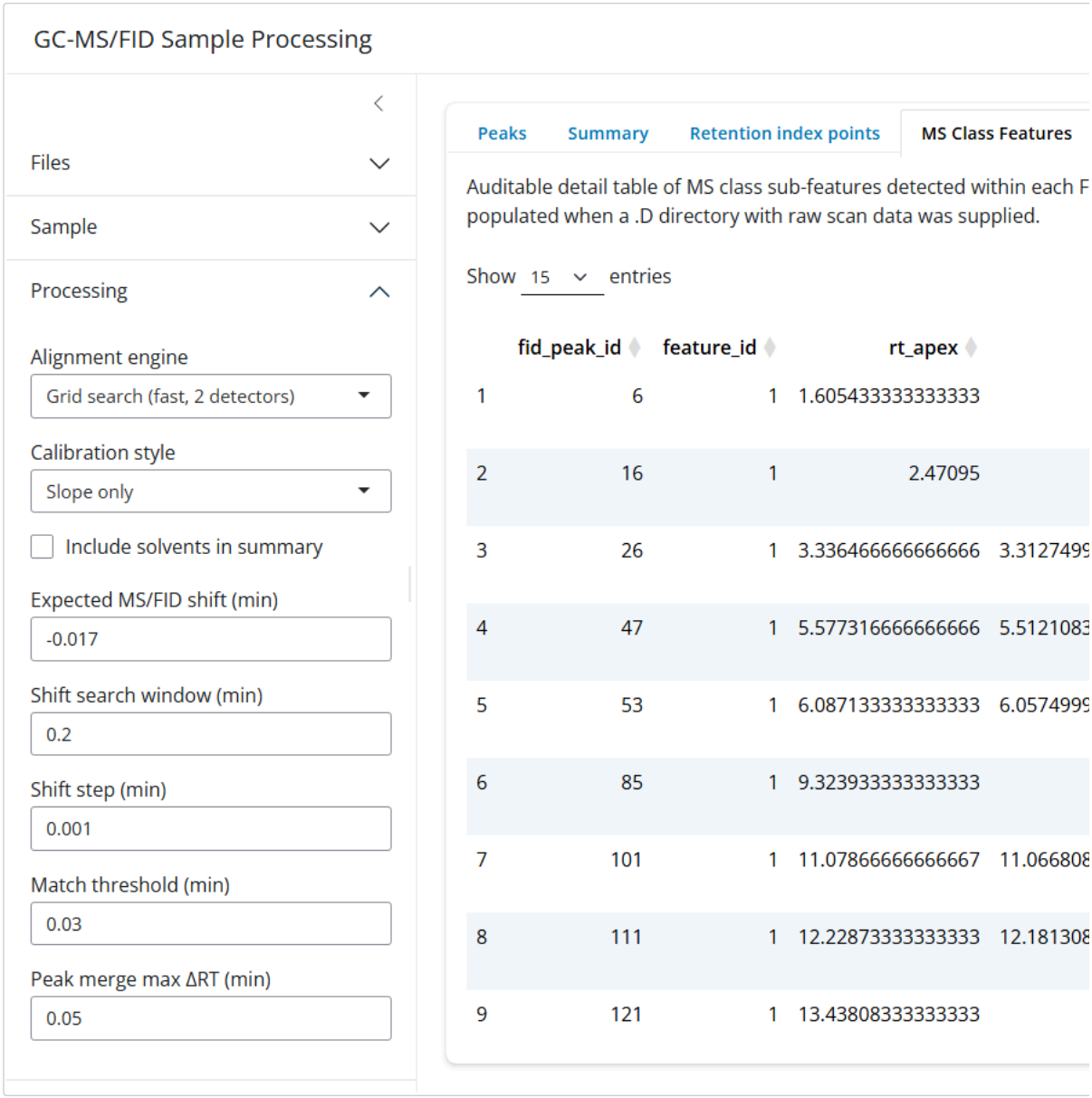
These are the numbers that go on the Summary Report. They match the code-based result because both paths call the same functions.

6. Retention-index anchors and class features



The RI points tab is editable; the polynomial is refit from whatever is there.

The retention-index model is refit at run time from these anchors, so editing a point changes how library hits are cross-checked.



The MS Class Features tab lists class sub-features detected inside each FID peak.

This audit table is only populated when raw scan data was supplied.

7. Download the reports

The **Excel report** and **PDF report** buttons call `write_excel_report()` and `write_pdf_report()` on the current result, named `{project code}_{sample number}_GC-MS-FID`.

Summary

The workflow is `prepare_peaks()`, then `calibrate()`, then `summarize_report()`, then a report writer; the app wraps the same sequence. The inputs to supply yourself are the sample metadata (density, methanol, water, dilution), the calibration style, and, for internal standardisation, the reference peak. Worth checking by eye: interpolated peaks, GC recovery, and the choice of alignment engine.

Screenshots are generated by `data-raw/capture_app_screenshots.R`; rerun it after any UI change.